When do kernel tests raise false alarms on long-memory data?

Exploratory manuscript draft

23 October 2026

Abstract

We make four contributions to the study of kernel tests on long-memory data. First, we measure false alarms under known null models. A test set to reject 5% of true nulls instead rejects 27.1% for split-sample maximum mean discrepancy (MMD) on stationary fractional Gaussian noise (fGn) at $H = 0.8$, $n = 2000$, with uncentred wild multipliers (271 of 1,000 series). The Hilbert–Schmidt independence criterion (HSIC) rejects 9.2% for independent fGn pairs at $H_X = H_Y = 0.7$, $n = 2000$, with subsampling (184 of 2,000 pairs). Long memory makes averages of kernel features fluctuate differently from short-memory averages; the tested critical-value rules capture different amounts of that dependence. Second, a Hermite-rank calculation explains why split MMD and independence tests have different memory rates. It identifies $H_X + H_Y = 1.5$ as a useful axis for the latter, without proving their critical-value laws. Third, a factorial experiment switches on heavy tails, nonlinear dynamics, and seasonality one at a time, with and without memory. Memory still drives most of the excess. A seasonal scale shared by two independent series is a separate failure that is specific to the kernel tests. With no memory and $n = 2000$, HSIC with wild multipliers rejects 73.3% of 1,000 such pairs, while Pearson correlation rejects 4.7% and HSIC circular shifts 5.0%. Fourth, we apply the rules to Nasdaq stock-trade durations. At $n = 2000$, HSIC shifts reject 3.4%–5.6% and subsampling 7.2%–10.2% across four 500-pair cells. These market rates are descriptive because process independence is unknown.

Keywords: long-range dependence; kernel tests; Hermite rank; MMD; HSIC; distance covariance; seasonality; market microstructure.

1 Introduction

This paper makes four contributions for machine-learning tests on ordered data. First, it measures false-alarm rates for split MMD, HSIC, and distance covariance when the null is known. At a nominal 5% level and $n = 2000$, split MMD rejects 27.1% of 1,000 stationary fGn series at $H = 0.8$ with uncentred wild multipliers. HSIC rejects 9.2% of 2,000 independent fGn pairs at $H_X = H_Y = 0.7$ with subsampling. Second, we use Hermite rank to explain the different memory rates of a one-series split and a product of two independent series. Third, we cross memory with heavy tails, nonlinear dynamics, and seasonality to ask which failures come from memory and which from other process features. Fourth, we examine the same independence rules on separated Nasdaq trade-duration windows, where the null cannot be verified.

MMD [Gretton et al., 2012] is often used to flag a change in distribution. HSIC [Gretton et al., 2007] is used to flag dependence between paired variables or representations. Distance covariance [Székely et al., 2007] asks the same question as HSIC through pairwise distances. All three tests were introduced for independent and identically distributed observations. A nominal 5% test should reject a true null about five times in 100 repeated samples. That interpretation becomes unreliable if the observations are persistent and the critical value does not reflect their dependence. The two rates above come from different nulls and different rules; they are examples of the problem, not a head-to-head ranking of MMD against HSIC.

Long memory means that correlation fades slowly with time. Even a long series may therefore contain much less information than the same number of independent observations. Kernel features inherit some of this persistence. Their averages can vary more than a critical-value rule based on short windows or short multiplier correlations expects. We vary memory, transformations, sample size, and rule to see where this mismatch appears.

The two tests require separate explanations. Split MMD compares the first and second halves of one stationary series; its null says the halves have the same distribution. HSIC and distance covariance compare two aligned series; their null says the two processes are independent, although each may have memory within itself. For a rank-one split-MMD feature, the mean can change rate as soon as $H > 1/2$. For the product of two independent rank-one features, the covariance calculation instead points to $H_X + H_Y = 3/2$. Neither calculation provides a critical value for the exact kernel tests used here.

The rule matters as much as the statistic. On the same independent-fGn cell, $(H_X, H_Y) = (0.7, 0.7)$ and $n = 2000$, HSIC rejects 9.2% with subsampling, 7.35% with wild multipliers, and 5.45% with circular shifts. The wild rule also rejects 6.5% at the $H_X = H_Y = 0.5$ baseline. Table 3 separates that baseline error from the change associated with stronger memory.

Memory is not the only process feature that can break these tests. We therefore switch on one feature at a time and cross it with memory. Heavy tails and nonlinear dynamics change the size of the memory effect but do not create a large excess on their own. Seasonality does. A seasonal mean shared by two independent series misleads every test we ran, including Pearson correlation. This is the classical nonsense-correlation problem [Yule, 1926]. A seasonal *scale* behaves differently. With no memory, it leaves Pearson at 4.7% but drives HSIC with wild multipliers to 73.3%. The kernel test sees that the two series swing widely at the same times, even though their signs are unrelated. This sensitivity to more than linear correlation is the reason to use a kernel test. On ordered data it is also an extra exposure.

The final application uses AAPL, MSFT, and NVDA trade durations to test the boundary of this account in market microstructure. Paired windows are at least 20 trading days apart, but that gap does not prove process independence. We treat their rejection fractions as descriptive observations, not measured false-alarm rates.

2 Theory and predictions

2.1 Memory and rank

Let ξ_t be a stationary Gaussian series. We write its autocovariance at lag k as

$$\rho(k) \sim ck^{-D}L(k), \qquad D = 2 - 2H, \quad 1/2 < H < 1, \tag{1}$$

where L varies slowly. The reference $H = 1/2$ has no long memory; larger H gives slower decay. Hermite rank m is the order of the first nonzero term when a centred feature $g(\xi_t)$ is expanded in Hermite polynomials. Its leading covariance has order $\rho(k)^m$. If $mD > 1$, that leading term is summable and suggests the usual square-root-n rate for a scalar partial sum. At equality, L can change the answer.

This calculation predicts different rates for different features. Rank one remains non-summable for every $H > 1/2$. Rank two reaches the balance $2D = 1$ at $H = 3/4$. It is a statement about transformed Gaussian averages, not a null law or critical value for MMD or HSIC. Long-memory two-sample U-process theory provides related results [Dehling et al., 2017]. A separate rank-two result for a Gaussian graphical-model statistic appears in Zhai et al. [2026]; its assumptions and statistic differ from ours.

2.2 Two-sample MMD

MMD asks whether two samples have the same distribution. Here they are the first and second halves of one observed series $Z_1, \ldots, Z_n$. We compare their kernel mean embeddings:

$$\mathrm{MMD}_n^2 = \left\| \frac{2}{n} \sum_{t \le n/2} \phi(Z_t) - \frac{2}{n} \sum_{t > n/2} \phi(Z_t) \right\|_{\mathcal{H}}^2 . \tag{2}$$

Stationarity makes the two population embeddings equal, so rejection under this generator is a false alarm. If the feature $\phi(G(\xi_t))$ has a nonzero rank-one term, its sample mean carries long memory for any $H > 1/2$. An even transform such as $G(\xi) = \xi^2 - 1$ can cancel that term and leave rank two as the leader. This is why we compare rank-one and even transforms. It predicts a difference in scale; it does not prove the null law of the implemented MMD statistic [Dehling et al., 2017].

2.3 Independence statistics

HSIC and distance covariance ask a different question: are the processes X and Y independent? Each process can still depend strongly on its own past. The leading cross-series term has the schematic form

$$U_t = \{f(X_t) - \mathbb{E}f(X_t)\}\{g(Y_t) - \mathbb{E}g(Y_t)\}. \tag{3}$$

If the two processes are independent and both factors have rank one, $\mathrm{Cov}(U_0, U_k)$ is the product of their within-series covariances. It decays at order $k^{-(D_X + D_Y)}$. The power-law balance lies at

$$D_X + D_Y = 1 \quad \Longleftrightarrow \quad H_X + H_Y = 3/2. \tag{4}$$

With equal memories, this corresponds to $H = 3/4$ in each series. We include unequal-memory pairs to see whether the sum organizes their rejection rates. The line is a covariance balance, not a universal point at which tests start to fail.

The closest published limit concerns a related statistic. Betken and Dehling [2026] prove a limit for the empirical joint characteristic-function process under independent Gaussian-subordinated series, with a distance-covariance corollary. On one side, their Gaussian conditions require *both* $D_X, D_Y > 1/2$, or both H values below $3/4$ [Betken and Dehling, 2026, Theorem 2.2(i) and Corollary 2.1(i)]. Their non-central result for $D_X + D_Y < 1$ uses Gaussian identity transforms [Betken and Dehling, 2026, Theorem 2.2(ii) and Corollary 2.1(ii)].

Those assumptions leave a mixed pair with one $H > 3/4$ but $H_X + H_Y < 3/2$ outside the stated result. The theorem also does not give a Gaussian-kernel HSIC limit. We therefore use Equation (4) to organize experiments, while keeping the proved distance-covariance case separate from the HSIC prediction.

Table 1: Scope of the theory used to motivate the experiments. No cited result validates every implemented critical-value rule.

Result	What it covers	What remains open here
Hermite covariance calculation	Rate of a scalar transformed Gaussian mean or independent product feature	RKHS-valued split-MMD and HSIC limits; critical-value laws
Betken and Dehling [2026], Thm. 2.2 and Cor. 2.1	Characteristic-function limit; distance-covariance corollary	Gaussian-kernel HSIC; mixed normal region; exact finite-sample rule below
Betken and Dehling [2026], Thm. 2.4	Subsampling for a statistic with a limit, under process and block conditions	Exact HSIC and dCov transfer; finite-sample size
Chwialkowski et al. [2014]	Wild bootstrap under weak-dependence assumptions	Long-memory validity of the present multiplier rule

Published critical-value results have their own entry conditions. The random-process HSIC test of Chwialkowski and Gretton [2014] and the distance-covariance block bootstrap of Betken et al. [2024] assume absolute regularity (β-mixing) at a polynomial rate. The RKHS self-normalized tests of Ghoshal and Shao [2026] assume summable strong-mixing coefficients; the authors note that this "requires the underlying temporal dependence to be of short-range type". Fractional Gaussian noise with $H > 1/2$ is not completely regular [Betken and Wendler, 2018, Example 1]. For a stationary Gaussian sequence, the maximal correlation is bounded by 2π times the strong-mixing coefficient [Bradley, 2005, Section 7], so such fGn is not strongly mixing and these conditions exclude our long-memory cells. The wild bootstrap of Chwialkowski et al. [2014] uses a τ-dependence coefficient. Taking one future observation and the identity as the one-Lipschitz test function gives $\tau(r) \geq \mathbb{E}|\mathbb{E}(Z_r \mid \mathcal{F}_0)|$. For a stationary Gaussian series, conditioning on the whole past leaves the conditional mean with variance at least $\rho(r)^2$, so $\tau(r) \geq (2/\pi)^{1/2}|\rho(r)|$. For fGn this bound decays like r^{2H-2}. Their rate conditions, $\sum_r r^2\tau(r)^{1/2} < \infty$ for the degenerate V-statistic limit and $\tau(r) = O(r^{-6-\epsilon})$ for MMD, therefore fail for every $H > 1/2$. Their MMD result also requires the two samples to be independent, which the two halves of one long-memory series are not. The HSIC wild bootstrap of Diz-Castro et al. [2026] assumes near-epoch dependence on a β-mixing process with summable rates. Its Gaussian limit has a covariance operator defined by a series over lags. For independent fGn pairs and a rank-one product feature, that series diverges when $H_X + H_Y \geq 3/2$ by Equation (4), so their conditions cannot hold there. Below that line, representing fGn through its own Gaussian innovations fails their rate condition, but we have not excluded every possible representation. Betken and Dehling [2026, Theorem 2.4] prove a subsampling result when a statistic has a nondegenerate limit and the process and block lengths meet their conditions. The theorem allows the offset $d_n = \lfloor 0.1n \rfloor$. Together with their distance-covariance limit, it may cover some independent-fGn dCov cases, but only after checking our exact statistic and block rule.

It does not validate our HSIC rule. An HSIC null limit under these assumptions is missing, and our subsampling windows reuse bandwidths selected from the full sample. Other long-memory subsampling and self-normalized block results also require statistic-specific conditions [Betken and Wendler, 2018, Bai et al., 2016]. Block choice can matter under Gaussian subordination [Zhang et al., 2022]. Table 1 separates these published results from what remains open here.

3 Experiments and critical-value rules

Every rule targets a nominal 5% rejection level. In a simulation cell, we know that the null is true: the independence arms generate independent processes, and the MMD arm splits one stationary process. Its rejection fraction is therefore an empirical false-alarm rate. We do not know whether the process-independence null holds for stock trades, so their fractions are *placebo rejection rates*. Wilson intervals measure counting uncertainty for simulation fractions, not uncertainty about the generator or the rule.

Table 2: Study arms. A simulation cell's null is fixed by its generator; the stock placebo has an unverified process-independence null.

Arm	Series and comparison	Sample size	Replications or pairs
Initial MMD	Stationary fGn and transforms, split in halves	500, 1,000, 2,000, 4,000	1,000–2,000
Initial independence	Independent fGn, subordinated and unequal-memory pairs	500, 1,000, 2,000	2,000 per cell
EXP1 H5/E2	MMD rule control and rank-one/two transforms	500, 1,000, 2,000	1,000–2,000
EXP1 E3	Ten equal or unequal independent fGn pairs	500, 1,000, 2,000	2,000 per cell
EXP2	Gaussian, heavy-tailed, nonlinear, and seasonal generators crossed with $H \in \{0.5, 0.7, 0.8\}$	500, 2,000	1,000 per cell
Stock placebo	Separated AAPL, MSFT, NVDA duration windows	500, 1,000, 2,000	500 pairs per cell
Revision control	Independent $H = 0.5$ fGn pairs; centred versus raw multipliers	500, 1,000, 2,000	1,000 per cell

We also compared centred and raw Markov multipliers on independent $H = 0.5$ Gaussian pairs. This control measures rule error before long memory is introduced. Its counts and selection rule are in Appendix A.

3.1 Initial MMD and independence experiments

The first MMD experiment splits each series in half. It uses fGn, the bounded transform $2\Phi(\xi)-1$, and $\operatorname{sign}(\xi)$. The fGn grid is $H \in \{0.5, 0.6, 0.7, 0.75, 0.8, 0.9\}$. We compare wild-multiplier and sampling-window critical-value rules.

We include a slow-mean-reversion comparison to distinguish persistent behaviour from a different dependence pattern. It follows $Y_t = (1-\lambda)Y_{t-1} + \varepsilon_t$, with fGn innovations at $H = 0.1$, $\lambda \in \{0.01, 0.1\}$, and a burn-in of $\lceil 10/\lambda \rceil$ observations. Generated copies are independent of one another, although innovations within a copy are correlated. This is a discrete mean-reverting model, not a claim of continuous-time fractional OU sampling.

The independence experiment compares Gaussian-kernel HSIC, distance covariance, and Pearson correlation as a linear reference. It uses independent fGn pairs, bounded and sign transforms, unequal-memory pairs, and independent copies of the mean-reverting process. Kernel sample sizes are $n \in \{500, 1000, 2000\}$; the Pearson reference uses $T \in \{500, 2000, 8000\}$.

The rules construct critical values in different ways. For HSIC and distance covariance, the wild rule uses 500 draws of empirically centred block-Markov Rademacher multipliers. Their correlation decays as $\exp(-k/\ell)$, with $\ell = \lfloor \sqrt{n} \rfloor$. The HSIC shift rule realigns one full series by circular offsets from $\lfloor 0.1n \rfloor$ to $n - \lfloor 0.1n \rfloor$. Subsampling instead measures contiguous windows of length $\ell = \lfloor \sqrt{n} \rfloor$, with a cross-series offset of $\lfloor 0.1n \rfloor$. Pearson uses Newey–West (NW) and equal-weighted-cosine (EWC) long-run variance rules. These settings describe the tested implementations; they are not general validity claims.

Here is the exact statistic used for HSIC. Let K^c and L^c be empirically double-centred Gaussian Gram matrices. The bandwidth for each is the median off-diagonal pairwise distance, with a positive-distance fallback for ties. We compute $T_n = n^{-1} \sum_{i,j} K^c_{ij} L^c_{ij}$. Distance covariance uses double-centred distance matrices in the same construction. In either case, $T_n = nV_n$ for the corresponding biased V-statistic.

For one multiplier draw w, the HSIC reference value is $T^*_n = n^{-1} w^\top (K^c \circ L^c) w$; distance covariance uses the analogous distance product. With B reference draws, the upper-tail p-value

is $(1+\#\{b : T_b^* \geq T_n\})/(B+1)$. The shift rule applies the same counting convention to zero-shift HSIC and the permitted circular alignments.

For subsampling, the full-series value is $T_n = nV_n$. Each window has its own value $T_\ell = \ell V_\ell$, and the rule rejects when T_n exceeds the empirical 95th percentile of the window values. The fGn generator is Davies–Harte. This specification fixes what was computed; it does not establish the rule's asymptotic validity.

3.2 Rule checks, rank change, and finer memory grid

The simulation extension has three parts. H5 compares five MMD multiplier variants at $H = 0.5$ to find rule error when long memory is absent. E2 tests the even rank-two transforms $\xi^2 - 1$ and $|\xi|$, adds $H = 0.55$, and compares multiplier lengths, two subsampling block rules, plug-in rate correction, and self-normalized MMD. E3 tests independence across a finer memory-sum grid.

Self-normalized MMD divides $n\,\mathrm{MMD}_n^2$ by a within-series measure of variation: $T_n^{\mathrm{SN}} = n\,\mathrm{MMD}_n^2\,/V_n$. The denominator V_n sums squared deviations of RKHS partial sums from straight-line bridges within each half, then divides by n^2. The critical value is the 95th percentile of this same ratio on every contiguous window of length $\lfloor n^{0.5} \rfloor$ or $\lfloor n^{0.7} \rfloor$. Under suitable conditions, the ratio can remove an unknown common scale. We do not assume those conditions validate this rule here. Appendix A gives the exact V_n.

E3 uses ten equal and unequal independent fGn pairs with $H_X + H_Y$ from 1.3 to 1.7. Each $n \in \{500, 1000, 2000\}$ cell has 2,000 replications. We plot the rates and changes between adjacent grid points; we do not fit a breakpoint.

3.3 Process features crossed with memory

EXP2 asks whether a test fails because of memory alone, a process feature alone, or the two together. Let ξ_t be unit-variance fGn with Hurst parameter H. At $H = 0.5$ it is independent standard Gaussian noise. Every generator is driven by this ξ_t, so the Gaussian row shares the same code path. We switch on one feature at a time:

- *G0, Gaussian baseline:* $X_t = \xi_t$.
- *HT-S, instantaneous heavy tails:* $X_t = F_{t_3}^{-1}\{\Phi(\xi_t)\}/\sqrt{3}$. This is an odd, increasing function of ξ_t, so it has Hermite rank one and the rank calculation of Section 2 applies.
- *HT-L, filtered heavy tails:* an ARFIMA$(0, H - 1/2, 0)$ filter of independent t_3 innovations. This process is not a function of a Gaussian series.
- *NL-SV, stochastic volatility:* $X_t = \exp(h_t/2)\varepsilon_t$, where h_t is a unit-variance AR(1) filter of ξ_t with coefficient 0.9 and ε_t is independent Gaussian noise. The series is uncorrelated, and its memory sits in its volatility.
- *NL-TAR, threshold autoregression:* $X_t = a(X_{t-1})X_{t-1} + \xi_t$, with $a(x) = 0.8$ for $x \leq 0$ and -0.4 otherwise.
- *SEAS-M, seasonal mean:* $X_t = \xi_t + \sin\{2\pi(t + U)/50\}$.
- *SEAS-V, seasonal scale:* $X_t = [1 + 0.5\sin\{2\pi(t + U)/50\}]\xi_t$.

The phase U is uniform on $\{0, \ldots, 49\}$ and is drawn afresh for each series. Each series in an independence pair therefore has its own phase, and the null of independent processes holds exactly. A diagnostic row gives both series the same fixed phase. These settings are illustrative and are not calibrated to market data.

Each generator is crossed with $H \in \{0.5, 0.7, 0.8\}$ and $n \in \{500, 2000\}$, with 1,000 replications per cell. The independence arm uses the HSIC, distance-covariance, and Pearson rules described above. The MMD arm uses the uncentred wild rule, square-root subsampling, and self-normalized blocks. We read two effects. The *alone* effect is the rate at $H = 0.5$ minus the Gaussian rate at $H = 0.5$. The *interaction* asks whether the rise from $H = 0.5$ to $H = 0.7$, and from $H = 0.5$ to $H = 0.8$, differs from the Gaussian rise. The hypotheses, generator checks, and a pilot were fixed in a design record before the main run.

3.4 Stock-duration placebo

The applied arm uses Databento XNAS.ITCH trade records[1] for AAPL, MSFT, and NVDA from January through June 2024. We keep trades from 09:30 to 16:00 New York time. Within each day, we sort by event timestamp and feed sequence, collapse prints with the same event timestamp, and measure the positive gaps between distinct timestamps. A gap is one intertrade duration.

We take the log of each duration in seconds and subtract that stock's pooled mean for its 15-minute clock bin. This removes a coarse time-of-day pattern. The local cache also keeps the raw gap, timestamp, day, and clock bin for checks described in Appendix A.

Each placebo pair contains two random-start windows of 500, 1,000, or 2,000 consecutive durations. A window stays within one day; the paired days are at least 20 trading sessions apart. We test AAPL–MSFT, AAPL–NVDA, MSFT–NVDA, and AAPL–AAPL, with 500 pairs per cell and 12 cells in total. The same-stock pair checks the sampling construction; it does not establish independence. We reuse the simulation's HSIC, distance-covariance, and Pearson implementations. Days can recur across pairs, and the time-of-day adjustment uses pooled data. These facts prevent us from treating the observed rejection fractions as known-null false-alarm rates.

4 Results

Table 3 gives the main comparison in one place. The $H_X = H_Y = 0.5$ column shows each rule's baseline error. The next two columns show rates as within-series memory strengthens. The change from $n = 500$ to $n = 2000$ shows that excess rejection does not necessarily grow over the sample sizes we tested.

Table 3: Independent-fGn false-alarm rates at nominal 5%. The first three numeric columns are percentages at $n = 2000$ for equal-memory pairs with $(H_X, H_Y) = (0.5, 0.5), (0.7, 0.7), (0.8, 0.8)$. Δ is the percentage-point change from $n = 500$ to $n = 2000$ at sum 1.6. Each cell has 2,000 replications. Theory status refers to the exact rule used here.

Rule	$H = .5$	Sum 1.4	Sum 1.6	Δ	What it limits	Theory status
HSIC wild	6.5	7.4	8.9	−2.4	Downweights lags on a $\sqrt{n}$ scale	Weak-dependence theory only
HSIC shift	5.0	5.5	5.7	−0.3	Excludes shifts $< 0.1n$; uses full series	No long-memory proof here
HSIC sub	5.6	9.2	19.7	−2.2	Uses only $\sqrt{n}$ windows	HSIC limit and rule match open
dCov wild	6.6	7.1	9.6	−2.2	Downweights lags on a $\sqrt{n}$ scale	Weak-dependence theory only
dCov sub	5.3	10.6	23.0	−2.2	Uses only $\sqrt{n}$ windows	Restricted dCov limit; exact rule open
Pearson NW	5.8	8.2	13.3	−0.3	Drops lags beyond fitted cutoff	Standard HAC excludes these cells
Pearson EWC	6.1	6.0	8.5	+0.8	Drops modes beyond $\lfloor 0.4n^{2/3} \rfloor$	Long-memory guarantee open

4.1 Split MMD: rank changes when the false alarms rise

The first MMD experiment shows an early increase for rank-one features. At a nominal 5% level, its rejection rate is about 12–19% by $H = 0.6$ and 44–71% by $H = 0.9$, depending on the

[1] Dataset and trade-schema documentation: https://databento.com/datasets/XNAS.ITCH and https://databento.com/docs/schemas/trades.

process and rule. This pattern agrees with the slow rank-one covariance decay. The tested grid does not locate a precise breakpoint.

We first checked the rule at the memory-free baseline. For Gaussian fGn at $H = 0.5$ and $n = 2000$, the old and published centred MMD multipliers reject 7.0% and 7.2% of true nulls. Independent and uncentred variants reject 5.3% and 5.0%. At $n = 500$, the old and centred rules reject about 9%. E2 therefore uses the uncentred rule that is closest to 5% in this control. This is a choice for the experiment, not a general claim that uncentred multipliers are better.

With that rule fixed, E2 compares feature ranks. At $n = 2000$, the even transform $\xi^2 - 1$ stays between 4.2% and 6.4% through $H = 0.75$, then rises to 11.3% at $H = 0.8$ and 22.8% at $H = 0.9$. The $|\xi|$ curve is similar. By contrast, the rank-one reference is already above nominal at $H = 0.55$ (Figure 1). Self-normalized block MMD varies less across the grid but still rejects about 6.0–8.5% on fGn and 4.7–7.5% on signs.

There is a cost to some apparent size improvements. In one $H = 0.7$ fGn shift alternative at $n = 2000$, self-normalized MMD detects the shift in 51% of replications. Conventional wild and subsampling rules detect it in 76–87% at their own cutoffs. Because their null rejection rates differ, this is not a size-adjusted power comparison. Plug-in rate correction is often conservative and also detects fewer alternatives in this cell.

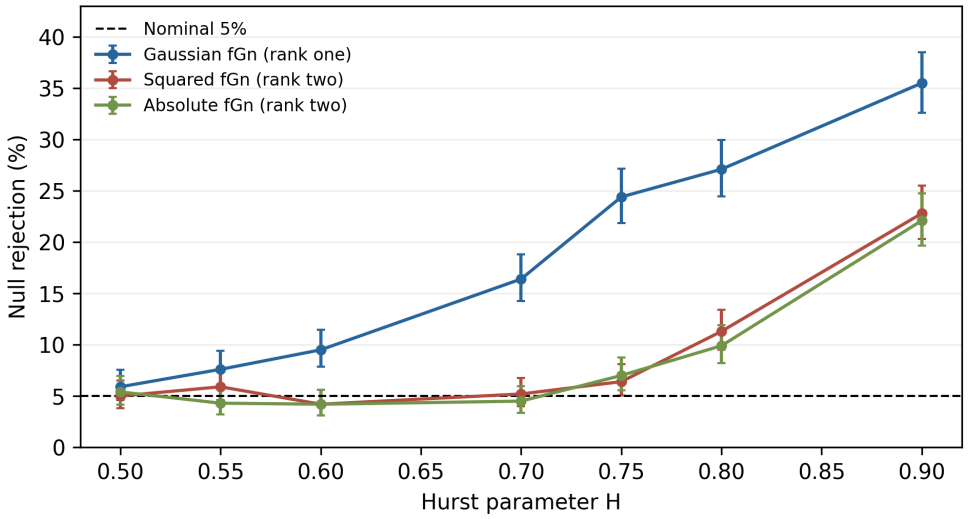


Figure 1: MMD rejection under the stationary split-series null at $n = 2000$, using the tested uncentred wild multiplier with length $\lfloor n^{0.5} \rfloor$. Points are saved E2 cell rates; bars are Wilson 95% counting intervals from 1,000 replications. The lines compare process-specific finite-sample behaviour, not proved asymptotic thresholds.

4.2 Independent pairs: the rule changes the answer

The independent-pair experiments show a different pattern. At $(H_X, H_Y) = (0.7, 0.7)$ and $n = 2000$, HSIC subsampling rejects 9.2% of true nulls and distance-covariance subsampling rejects 10.6%. On those same pairs, HSIC rejects 7.35% with wild multipliers and 5.45% with circular shifts. Pearson with NW and EWC references rejects 8.15% and 5.95%. Thus the critical-value rule changes the measured false-alarm rate even though the data and statistic are fixed. Oversized distance-covariance subsampling is not new. Betken and Dehling [2026, Appendix C, Table 1] report empirical sizes of 7.3–13.1% at $H = 0.6$ and 11.2–17.5% at $H = 0.7$ for a lag-weighted dCov statistic on independent fGn pairs, with $n \leq 1000$ and blocks of length $n^{0.4}$ or $n^{0.5}$. Our dCov subsampling rows extend that observation to $n = 2000$ and place it beside the wild and shift rules and the Gaussian-kernel HSIC statistic.

This excess appears below the covariance balance $H_X + H_Y = 1.5$. HSIC subsampling stays near 9.2% across $n = 500, 1000, 2000$ (9.1%, 9.15%, 9.2%). The wild-multiplier rule rejects 6.45% even at the $H_X = H_Y = 0.5$ baseline and $n = 2000$. Its excess at sum 1.4 therefore includes an error already present without long memory. At higher memory, subsampling becomes much more oversized, while circular shifts stay closest to 5% on this grid (Table 3). None of these patterns proves a rule valid outside the tested cells.

The unequal pair $(0.55, 0.85)$ has rates similar to the equal pair $(0.7, 0.7)$. Both have memory sum 1.4, so the sum is useful for organizing this grid. Figure 2 shows the rates against that sum; Figure 3 shows changes between adjacent grid points. The algebraic line at 1.5 helps read the plots, but the plots do not locate a common rejection boundary or prove a mixed-memory limit for these tests.

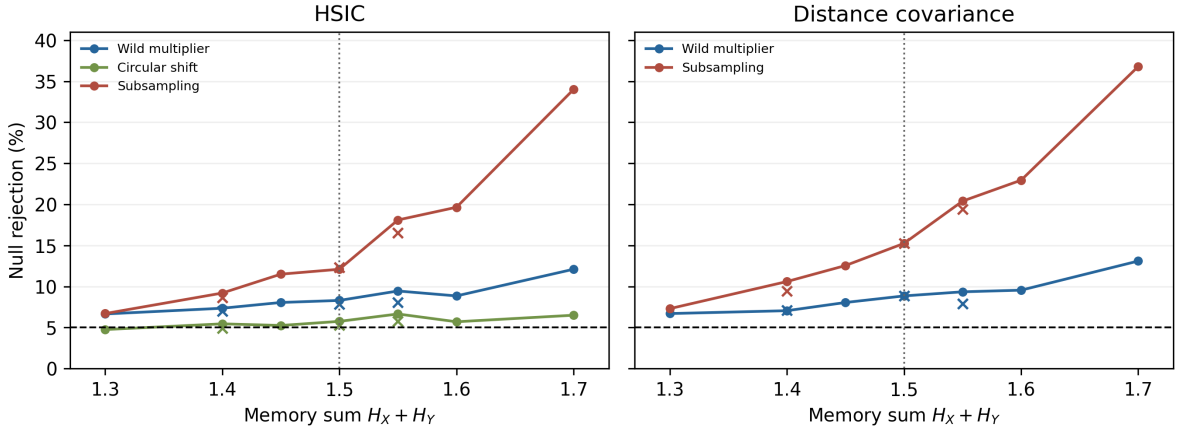


Figure 2: E3 null rejection at $n = 2000$ for independent fGn pairs. Connected circles are equal-memory pairs; crosses are unequal pairs. The horizontal dashed line is the nominal 5% rate and the vertical dotted line marks the candidate sum $H_X + H_Y = 1.5$. Each point has 2,000 replications. No line is a proved rejection boundary.

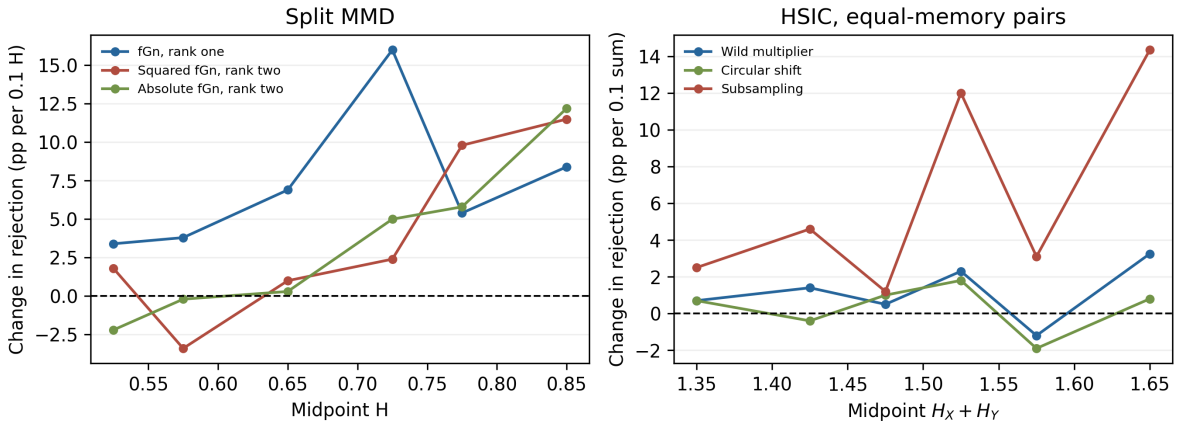


Figure 3: Adjacent-grid changes in null rejection at $n = 2000$. Left: split MMD under the uncentred wild rule, measured in percentage points per 0.1 increase in H. Right: HSIC on equal-memory independent pairs, measured per 0.1 increase in $H_X + H_Y$. Points sit at interval midpoints. These finite differences are descriptive, depend on grid spacing and Monte Carlo error, and do not estimate a change point.

We also tested whether raw multipliers would fix the memory-free HSIC and distance-

covariance baseline. They moved rejection from above 5% to below it and did not improve every cell. We therefore kept the original centred rule for the stock placebo. Appendix A gives the paired counts and selection rule.

4.3 Process features: seasonal scale is a kernel-specific failure

Table 4 gives the independence results of EXP2 at $n = 2000$. Four findings stand out. Monte Carlo standard errors are about 0.7 points near 5% and 1.6 points near 50%. We correct the 288 tests behind the EXP2 verdicts with Holm's step-down method. These are 96 one-sided tests of an effect without memory and 192 two-sided tests of a change in the memory effect. Holm's method controls the familywise error under any dependence between the tests. Claims that compare rejection levels at $H = 0.8$ are corrected in a larger family, which adds 96 level comparisons for 384 tests in all. Every claim below survives this correction unless we say otherwise.

Table 4: EXP2 null rejection percentages for independent pairs at $n = 2000$ and nominal 5%, with $H_X = H_Y = H$. Each cell has 1,000 replications. The last row is a diagnostic in which both series share one fixed seasonal phase. It is excluded from the effect estimates.

	HSIC wild		HSIC shift		HSIC sub		dCov sub		Pearson NW	
Generator	.5	.8	.5	.8	.5	.8	.5	.8	.5	.8
G0 Gaussian	6.8	8.1	5.3	5.3	5.8	17.5	6.8	22.8	5.0	14.1
HT-S tails, instantaneous	6.1	7.0	4.5	3.8	4.3	15.7	3.2	16.5	5.7	11.5
HT-L tails, filtered	8.9	10.6	6.4	5.3	6.8	25.9	3.2	25.9	4.2	16.2
NL-SV volatility	7.9	6.7	5.8	3.9	4.9	6.2	2.2	1.9	5.2	4.6
NL-TAR threshold AR	7.1	11.9	4.7	7.0	7.2	29.8	6.4	33.5	6.6	12.1
SEAS-M seasonal mean	91.2	85.2	5.8	5.9	90.9	85.4	91.3	86.6	91.4	84.5
SEAS-V seasonal scale	73.3	68.0	5.0	5.1	71.5	73.0	45.6	53.3	4.7	13.4
Shared fixed phase	100.0	100.0	34.3	27.9	100.0	100.0	100.0	100.0	100.0	100.0

First, a seasonal mean misleads every single-path test, kernel or linear. With no memory, HSIC, distance covariance, and Pearson each reject about 91% of independent pairs. The phases are independent, so the processes are independent. Within one path, however, each series carries a fixed sinusoid with the same period. The sample correlation of two such sinusoids depends on their phase gap and does not shrink as n grows. The random phase acts as a component that never forgets, so a single path cannot average it out. This is the nonsense-correlation problem of Yule [1926], not a kernel-specific failure. Circular shifts stay at 5.8%. A shift moves one series against the other and re-randomizes their phase gap, which is what the null requires. Small declines at $H = 0.8$ in these rows reflect a ceiling near 90%, not a protective effect of memory.

Second, a seasonal scale misleads the kernel tests but not Pearson. With no memory, HSIC rejects 73.3% with wild multipliers and 71.5% with subsampling, and distance covariance rejects 45.6% with subsampling. Pearson-NW rejects 4.7%. A shared seasonal scale makes $|X_t|$ and $|Y_t|$ large at the same times while their signs stay unrelated. Pearson correlation sees only the linear layer, which is unaffected. HSIC and distance covariance also see the squared layer. SEAS-V is not a function of a Gaussian series, so the Hermite calculation of Section 2 does not apply directly. Informally, the shared pattern lives in the squared features of the two series, the analogue of the second Hermite layer that the even transforms isolate in the MMD experiment. Circular shifts again stay at 5.0%. When both series share one fixed phase, even circular shifts reject 28–34%, because no shift can break a common deterministic pattern.

Third, heavy tails matter little. The instantaneous t_3 transform tracks the Gaussian row within noise, as its rank-one structure predicts. The filtered t_3 process raises HSIC subsampling to 25.9% at $H = 0.8$, against 17.5% for the Gaussian row (Holm $p \leq 0.002$). Its gain over the Gaussian memory effect, 7.4 points, does not survive correction (Holm $p = 0.12$). For

distance-covariance subsampling, the filtered process is within noise of the Gaussian row at $H = 0.8$.

Fourth, the two nonlinear processes move the memory effect in opposite directions. Stochastic volatility places the memory in the variance of each series. HSIC and distance-covariance subsampling then stay at or below 6.2% even at $H = 0.8$, against 17.5% and 22.8% for the Gaussian row. The threshold autoregression adds 11–12 points to both subsampling rules at $H = 0.8$. That process also adds short-range persistence of its own. Its mean local-Whittle memory estimate is 0.59 at nominal $H = 0.5$ and $n = 2000$, against 0.50 for the Gaussian row. Its excess is therefore not a pure nonlinearity effect.

The MMD arm agrees with this picture. Uncentred wild MMD rejects 26–31% at $H = 0.8$ across the five non-seasonal generators, including 26.5% for the Gaussian row. Stochastic volatility breaks square-root subsampling even without memory (13.3% at $H = 0.5$), because its AR(0.9) volatility persists beyond a $\sqrt{n}$ window. Self-normalized blocks stay between 3.1% and 13.0% across the non-seasonal generators and memory levels. Seasonal series make split MMD conservative (0–2% at $H = 0.5$), because both halves contain whole seasonal cycles.

4.4 Stock durations: an applied boundary test

The six-month extraction gives 124 trading days per stock and 9.68 million AAPL, 6.57 million MSFT, and 14.21 million NVDA positive durations. Every stock has observations in all 26 clock bins. The paired windows cannot overlap in calendar time because they come from separated days. Separation alone does not make the underlying processes independent.

At $n = 2000$, HSIC circular shifts reject 3.4%–5.6% of the sampled pairs, wild multipliers reject 4.8%–8.6%, and subsampling rejects 7.2%–10.2% (Table 5). The table gives each stock pairing, including the AAPL–AAPL check. At $n = 500$, wild-multiplier rates are 8.6%–11.0% and subsampling rates are 9.4%–11.6%. There is no consistent rise with window size across these cells.

Table 5: Stock duration placebo rejection percentages at $n = 2000$ and a nominal 5% level. Each row has 500 pairs. The processes' independence is unverified and days are reused, so these are descriptive fractions rather than false-alarm estimates.

Pairing	HSIC wild	HSIC shift	HSIC sub	dCov wild	dCov sub
AAPL–MSFT	8.6	5.6	8.4	8.4	8.8
AAPL–NVDA	5.8	4.8	9.2	6.0	7.2
MSFT–NVDA	7.2	5.4	10.2	6.4	11.2
AAPL–AAPL	4.8	3.4	7.2	5.0	7.6

The trade records also limit what we can test. On a typical day, about one fifth of AAPL trades, one quarter of MSFT trades, and more than one third of NVDA trades have unspecified aggressor side, so we do not analyze trade signs. Days recur across sampled pairs; one day is used as many as 9–12 times on one side of a cell. Market regimes may persist across the 20-session gap. We cannot interpret Table 5 as a causal effect or a measured 5% false-alarm check.

5 Discussion

The simulations establish rule-specific false-alarm rates under their stated null generators. They show excess rejection at a memory sum of 1.4, below the covariance balance at 1.5. That balance predicts a change in the summability of a leading product feature; it does not determine a usable critical value. Circular shifts are closest to 5% on this grid, while subsampling is strongly oversized in several cells. We cannot extend either observation to other long-memory processes or changing market conditions without further work.

Theory and experiment meet only within stated assumptions. The subsampling theorem discussed in Section 2 may apply to restricted independent-fGn distance-covariance settings once our statistic and block rule are checked exactly. We still lack the corresponding HSIC null limit and a proof for its window rule. The MMD feature-rank argument likewise does not validate its multiplier or self-normalized critical values.

The stock study shows how the rules behave on observed market durations, where a true process-independence null is unavailable. The 20-session gap prevents overlap but cannot turn a rejection into a measured false alarm. A stronger claim would require a specified null, a validated rule, and matched surrogates. EXP2 separates memory from heavy tails, volatility, nonlinear dynamics, and seasonality in simulation, and its seasonal-scale result bears on the stock arm. We removed a clock-bin mean from log durations but not a clock-bin scale. A window of 2,000 durations covers about 7 to 15 minutes of a 390-minute session, so it sees only a small part of the daily cycle. Whether a residual intraday scale pattern contributes to the stock fractions is untested.

EXP2 separates two failures that a user would see as the same false alarm. The first comes from memory. Process features change its size, but not its direction, except stochastic volatility, which moves memory into a layer the independence tests barely weight. The second comes from seasonality. A shared seasonal mean breaks every single-path test, so its remedy is deseasonalization or a shift-based rule, not a different kernel. A shared seasonal scale breaks HSIC and distance covariance while leaving Pearson at 5%. That failure is specific to tests that look beyond linear correlation. On this grid, circular shifts were the only rule that held size under both seasonal forms.

Kernel choice is a second lever, alongside the critical-value rule. The E2 results suggest that features without a rank-one component hold size until $H = 3/4$. A kernel or feature map built to remove the first Hermite layer might therefore postpone over-rejection. The same layer, however, carries the first-order effect of a mean shift. For a feature g of a standard Gaussian ξ, $\partial_\mu \mathbb{E}g(\xi + \mu)|_{\mu=0} = \mathbb{E}\{\xi g(\xi)\}$, which is the first Hermite coefficient of g. A feature with no rank-one component therefore responds to a small mean shift only at second order. Whether a kernel can trade some of this sensitivity for size control, and at what cost in power, is open. Answering it requires size and size-adjusted power on the same cells, which this paper does not report.

Three proof problems remain. First, HSIC needs a null limit that covers transformed features and mixed-memory pairs. Second, an H-free MMD rule needs a specified statistic and block construction, a size proof, and a power comparison. Third, any claim that local asymmetry changes Hermite rank needs an explicit transform, kernel, and asymmetry sequence. These are open questions, not results of this paper.

6 Limitations

Our main simulation map covers Gaussian and Gaussian-subordinated series and one discrete slow-mean-reversion comparison. EXP2 adds four processes outside the Hermite framework, but at one parameter setting each, with 1,000 replications per cell and $n \leq 2000$. Its seasonal generators use one period, one amplitude, and a uniform random phase. Kernel tests stop at $n = 2000$. Flat rates over this range do not tell us what happens as n becomes much larger. Adjacent-grid slopes are noisy finite differences, not estimates of a structural threshold. We test one main bandwidth rule and only a small set of power alternatives. Some rules already miss 5% at the $H = 0.5$ baseline.

The market illustration covers three stocks and six months. Reused days and changing stock-specific regimes make its pair fractions dependent. Pooled clock-bin adjustment may leave within-bin seasonality and day-to-day shifts. Raw gaps and timestamps remain in the local cache for further checks. No theorem in Section 2 establishes validity for the exact implemented

HSIC or MMD rules or for the stock-window comparison.

7 Conclusion

Under the simulated nulls, rank-one split MMD begins to over-reject at weaker memory than the even rank-two transforms. For independent pairs, HSIC and distance-covariance rejection depends strongly on the critical-value rule, even below a memory sum of 1.5. A seasonal scale shared by two independent series is a second, kernel-specific source of false alarms that appears without any memory. Circular shifts resisted it on our grid. Hermite rank explains why these patterns differ, but it does not supply a valid cutoff. The stock results show the behaviour of the rules on market data; without a known independence null, they remain descriptive.

Data and code availability. The public repository and reproduction paths are given in Appendix A. It contains simulation code and outputs, stock request manifests and derived results, and the manuscript source. Raw Databento trades are excluded. No market-data redistribution right is claimed.

Ethics, funding, and competing interests. This draft reports simulation and market-trade analyses, with no human-subject intervention. Funding, author contributions, AI-tool use, and competing-interest statements require author verification before submission.

A Reproducibility and audit trail

The public GitHub repository contains the code and saved outputs named below. All paths are relative to its root. Raw Databento files are excluded and require separate licensed access. The stock manifests give source-file hashes, dates, and API cost quotes. The stock experiment README gives the commands to download, extract, test, and summarize those data.

Simulation outputs. The initial experiments are in two `experiments/` folders ending `mmd-dml` and `kernel-linear`. The `EXP1-20261016` folder holds the H5, E2, and E3 scripts, cell files, and summaries. Its manifests record seeds and target counts. The `EXP2-20261023` folder holds the process-feature design record, generator checks, pilot, main-run cell files, `summary_main.csv`, and `RESULTS.md`. The design record was written before the pilot and lists two later amendments.

The initial MMD cells used 2,000 null replications up to $n = 2000$ and 1,000 at $n = 4000$ after a runtime fallback. E2 used 2,000 null replications up to $n = 1000$ and 1,000 at $n = 2000$; each power cell used 500. E3 used 2,000 per cell. The paper's `make_figures.py` reads the saved E2/E3 summaries and records input and output hashes in `figures/manifest.json`. It does not rerun simulations.

Self-normalized block statistic. Let $S_{a:b} = \sum_{t=a}^{b} \phi(Z_t)$ and $m = \lfloor n/2 \rfloor$. The denominator in `mmdcal.py` measures how far feature partial sums deviate from a straight-line bridge within each half:

$$V_n = \frac{1}{n^2} \left\{ \sum_{k=1}^{m} \left\| S_{1:k} - \frac{k}{m} S_{1:m} \right\|^2 + \sum_{j=1}^{n-m} \left\| S_{n-j+1:n} - \frac{j}{n-m} S_{m+1:n} \right\|^2 \right\}. \tag{5}$$

The test uses $n\,\mathrm{MMD}_n^2 / V_n$. To find a critical value, it computes the same ratio in every contiguous block of length $\lfloor n^{0.5} \rfloor$ or $\lfloor n^{0.7} \rfloor$ and takes the empirical 95th percentile. Each block is split into halves. A change in block length changes the rule. Self-normalization alone does not prove that either choice has the intended size.

Multiplier control and decision record. The $H = 0.5$ control is in the repository's `investigations` directory; the reproduction guide gives its full path. At each sample size, centred and raw Markov multipliers were applied to the same 1,000 independent Gaussian pairs, with 500 multiplier draws per pair.

The selection rule required raw-multiplier rejection between 3% and 7% in all six test-by-size cells. It also required every raw-multiplier cell to be at least as close to 5% as its centred counterpart. Raw multipliers rejected below 5% in all six cells and failed the second condition in two. The stock arm therefore retained the centred implementation. The interpretation decisions dated 18, 19, and 22 October are in `FINDINGS.md`; they are not validation results.

Stock extraction and pair check. `experiments/STOCK-20261023/` contains request manifests, an extraction summary, code, 12 pair CSVs, and `stock_placebo_summary.json`. The summary script checks pair IDs, test status, day gaps, and within-day boundaries. All 6,000 saved pairs passed those checks; none had fewer than 20 trading sessions between windows. Raw trades and local duration caches are excluded from version control. The pair results remain descriptive because process independence is unknown and days recur across pairs.

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